

Tensor model of the psyche

based on social evidence and resource constraints

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<https://www.nsu.ru>

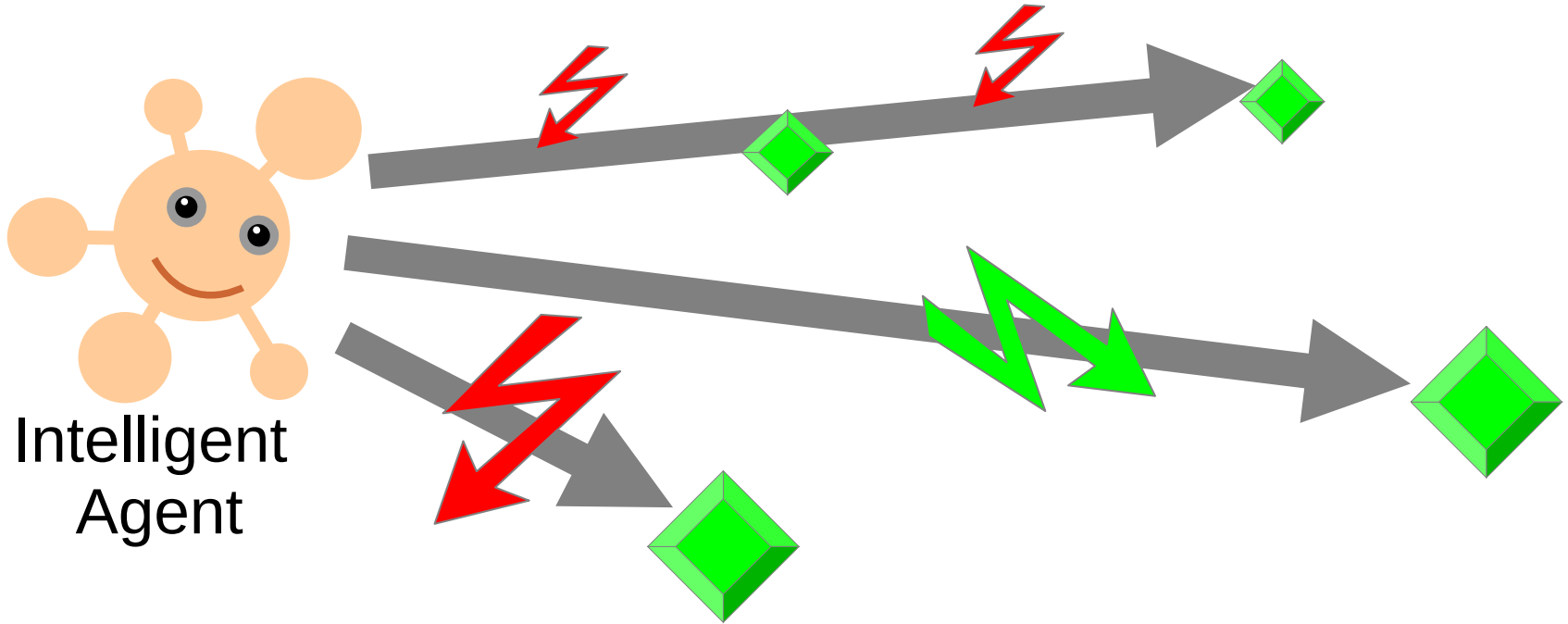


<https://agirussia.org>

General Intelligence:

Reaching complex goals **in different** complex environments, **using limited resources**

Ben Goertzel + Pei Wang + Shane Legg + Marcus Hutter

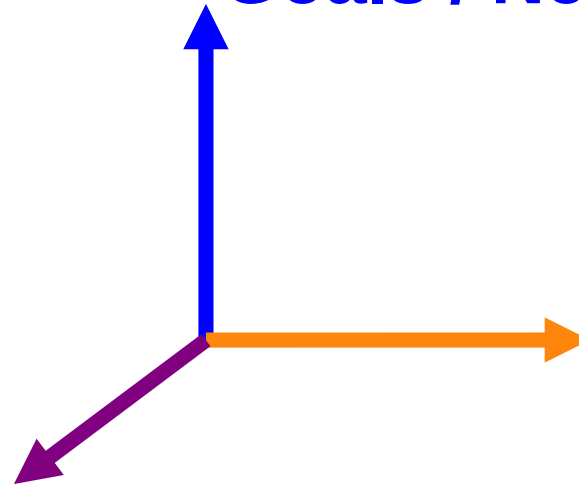


General Intelligence:

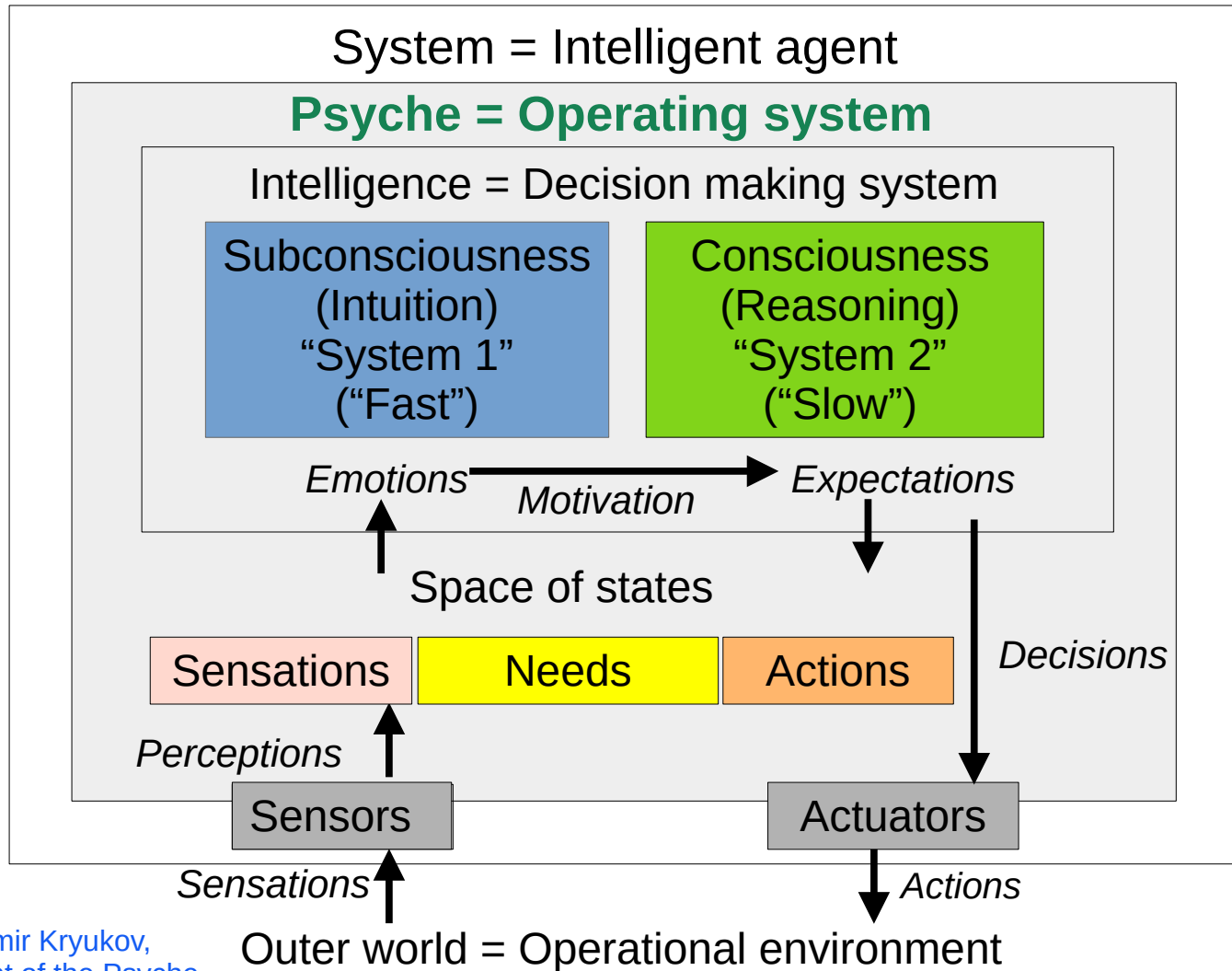
Reaching complex goals in different complex environments, using limited resources by means of complex actions

Goals / Needs

Actions



Signals from Environment / Sensations



Psyche = Operating system

Intelligence = Decision making system

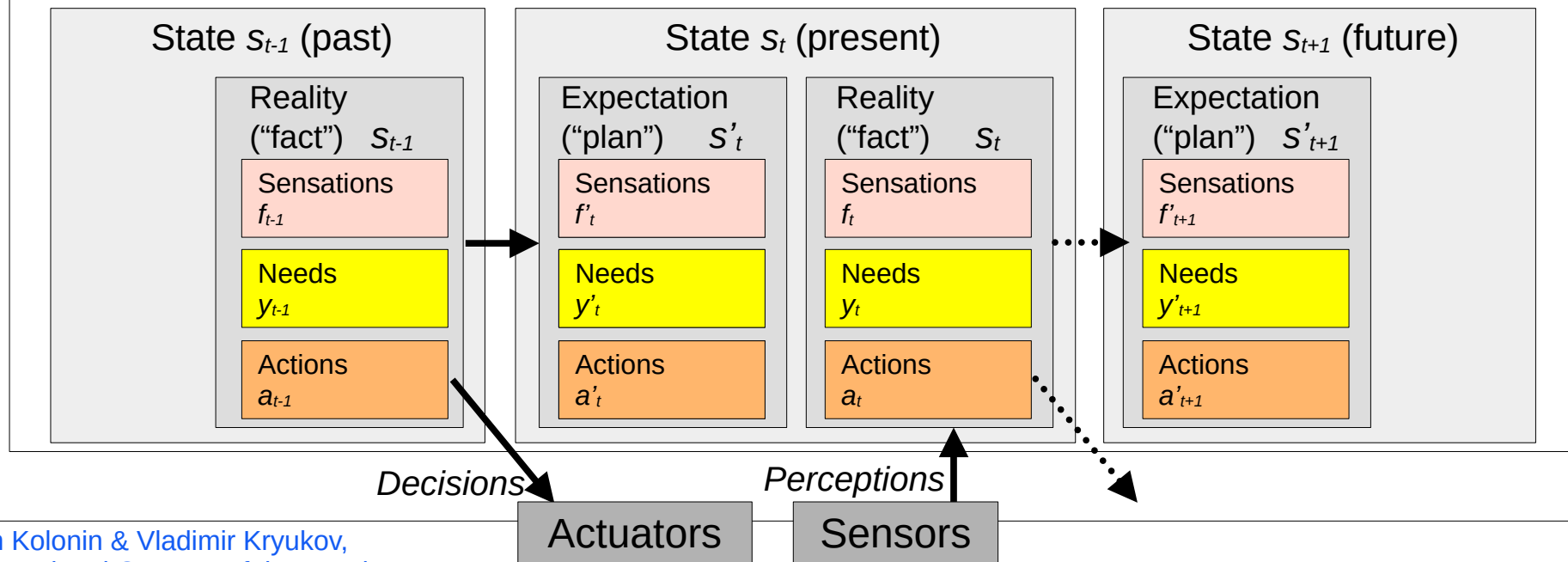
Models s ("invariants") of states with utilities U and probabilities P of transitions

$$U(\{s_t\}_{t \in [-T, -1]}, s'_0) = L(x \cdot (y_t - y_{t-1}), (s'_t - s_t), E(a_{t-1})) \quad s'_t = \operatorname{argmax}_s (U(\{s_t\}_{t \in [-T, t-1]}, s'_t), P(\{s_t\}_{t \in [-T, t-1]}, s'_t))$$

↑ Experiential learning

↓ Decision making

Space of states and episodic memory ("precedents")



Psyche = Operating system

Intelligence = Decision making system

Models s ("invariants") of states with utilities U and probabilities P of transitions

$$U(\{s_t\}_{t \in \{-T, -1\}}, s'_0) = L(x \cdot (y_t - y_{t-1}), (s'_t - s_t), E(a_{t-1})) \quad s'_t = \operatorname{argmax}_s (U(\{s_t\}_{t \in \{-T, t-1\}}, s'_t), P(\{s_t\}_{t \in \{-T, t-1\}}, s'_t))$$

↑ Experiential learning

↓ Decision making

Space of states and episodic memory ("precedents")

State s_{t-1} (past)

Reality
("fact") s_{t-1}

Sensations
 f_{t-1}

Needs
 y_{t-1}

Actions
 a_{t-1}

State s_t (present)

Expectation
("plan") s'_t

Sensations
 f'_t

Needs
 y'_t

Actions
 a'_t

Reality
("fact") s_t

Sensations
 f_t

Needs
 y_t

Actions
 a_t

State s_{t+1} (future)

Expectation
("plan") s'_{t+1}

Sensations
 f'_{t+1}

Needs
 y'_{t+1}

Actions
 a'_{t+1}

Decisions

Perceptions

Actuators

Sensors

$x \cdot y_t$ — "motivation vector"

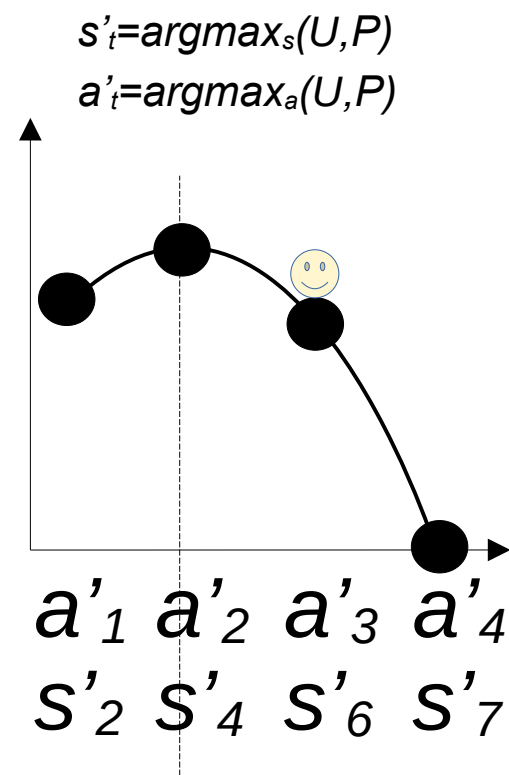
V. F. Petrenko and A. P. Suprun, "Goal oriented systems, evolution, and the subjective aspect in systemology," Tr. Inst. Sist. em. Analiza RAN 62 (1) (2012)

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E.E. Vityaev, A.V. Demin, Cognitive architecture based on the functional systems theory, 2018
<https://doi.org/10.1016/j.procs.2018.11.072>

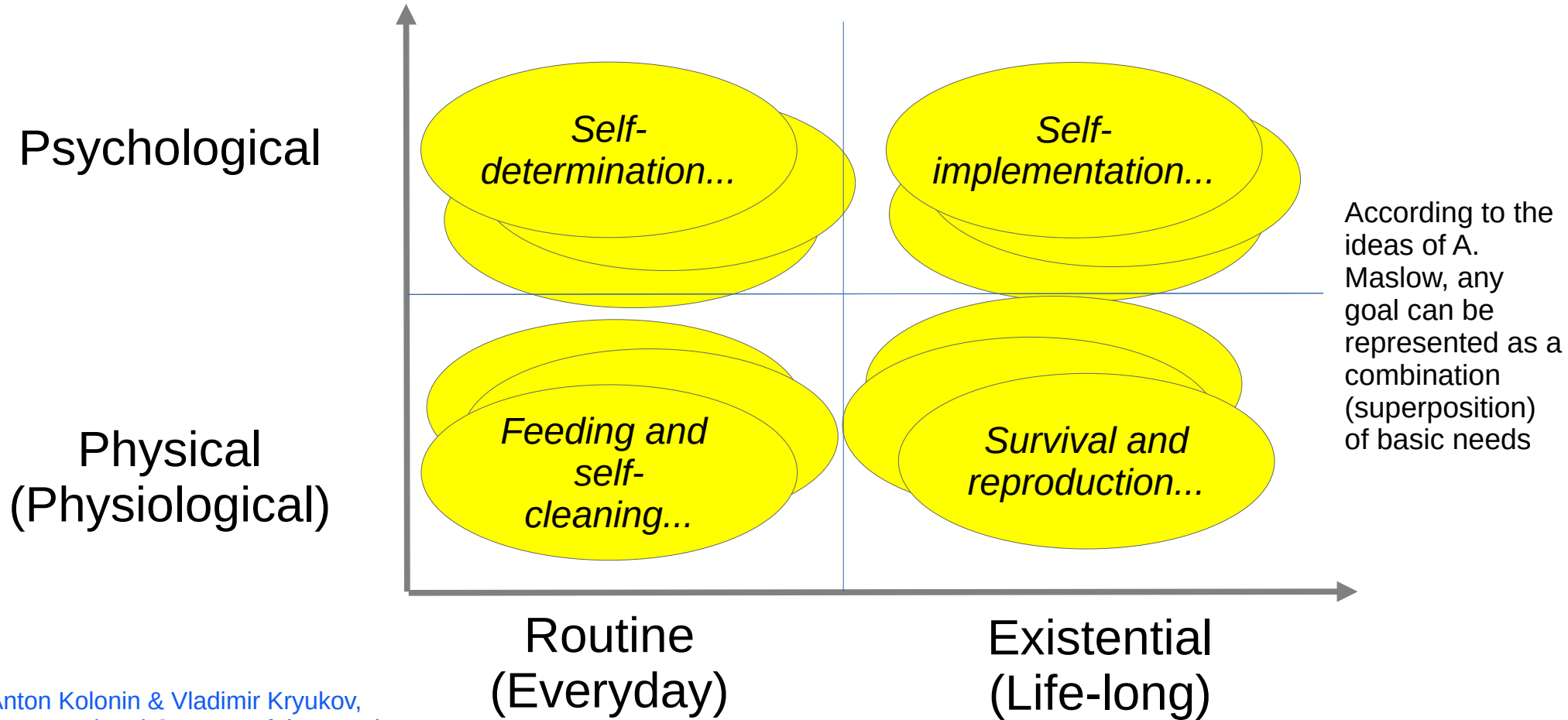
Decision making as operational risk management

S_t	S'_{t+1}	S'_{t+1}			U	P	$\Sigma U*P$
		a'	y'	f'			
S_1	S'_2	a'_1	y'_1	...	1.0	0.5	<u>0.7</u>
S_1	S'_3	a'_1	y'_2	...	0.4	0.5	
S_1	S'_4	a'_2	y'_3	...	1.0	0.8	<u>0.8</u>
S_1	S'_5	a'_2	y'_4	...	0.0	0.2	
S_1	S'_6	a'_3	y'_5	...	0.6	1.0	<u>0.6</u>
S_1	S'_7	a'_4	y'_6	...	0.0	1.0	<u>0.0</u>

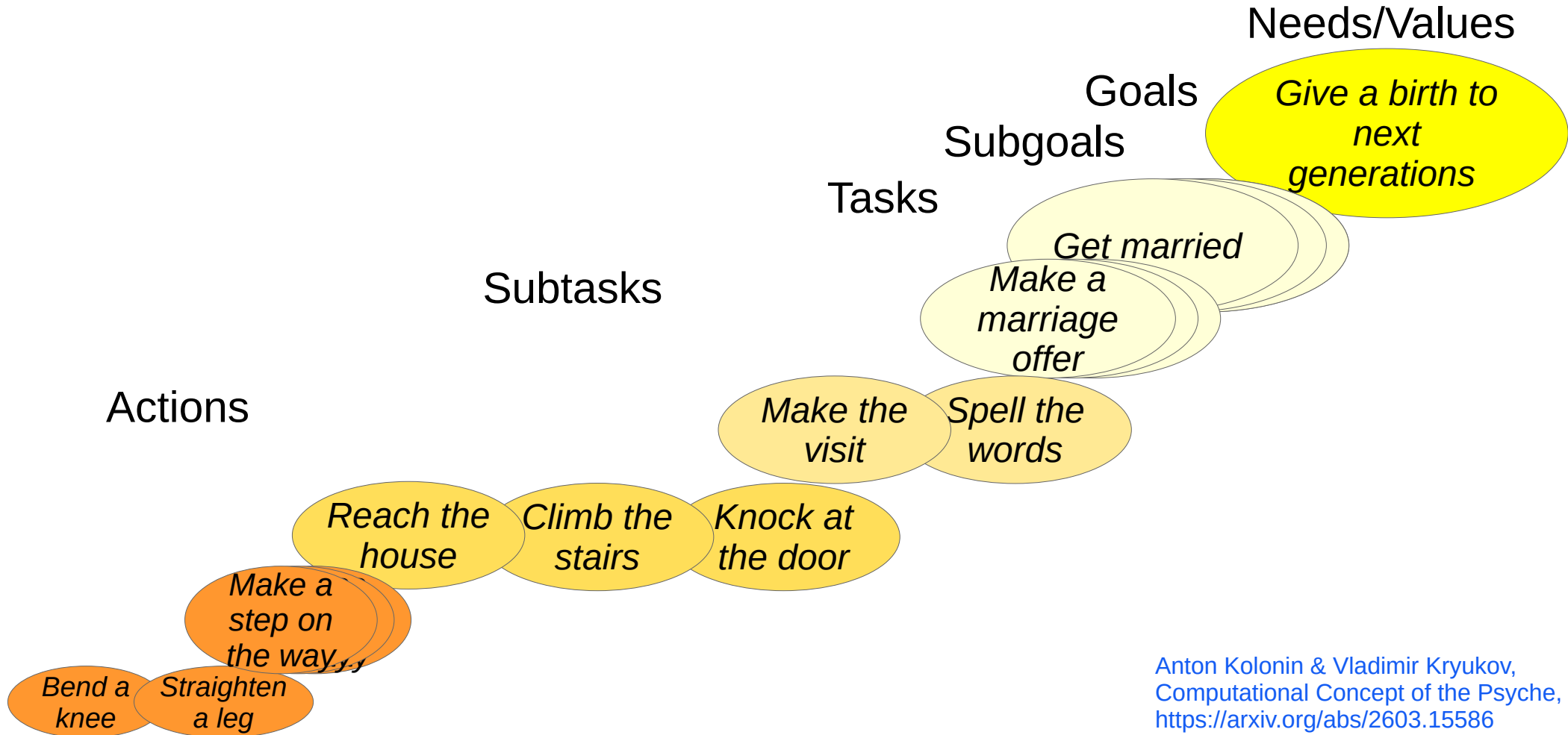


Tversky & Kahneman:
 most people choose a'_3 и s'_6
 ("smaller profit with
 greater reliability")

Space of Needs/Goals/Values



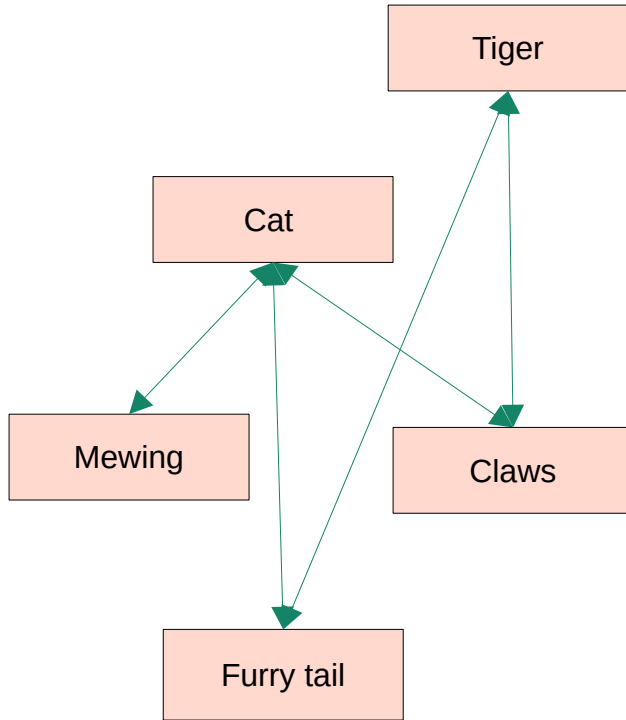
Hierarchy of values/goals/subgoals/tasks/subtasks



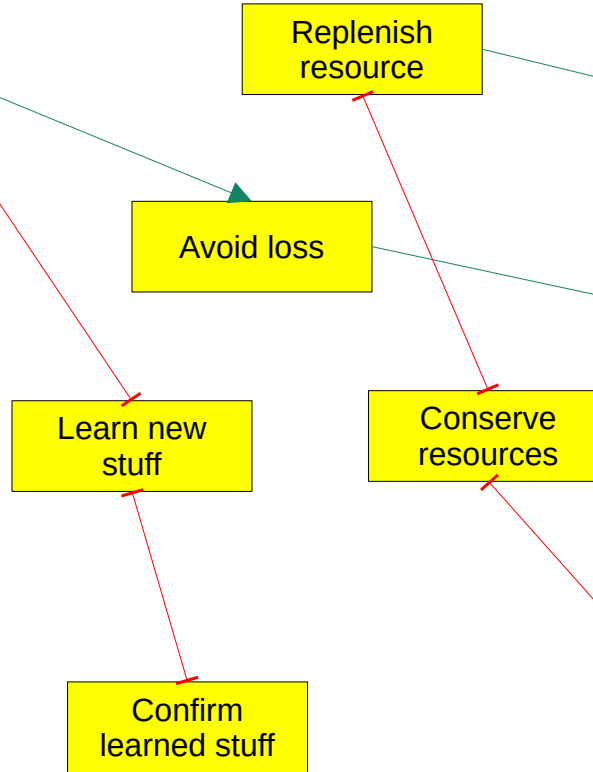
Anton Kolonin & Vladimir Kryukov,
Computational Concept of the Psyche,
<https://arxiv.org/abs/2603.15586>

Mutual dependency of state variable subgraphs tensors

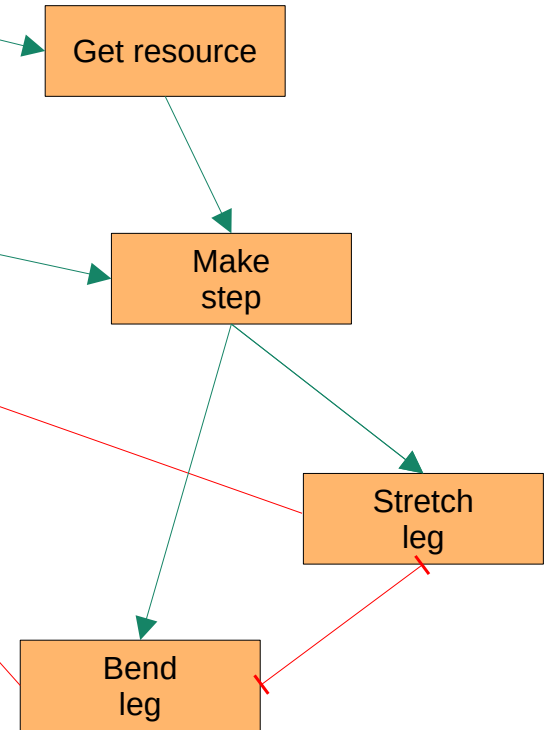
Percept feelings



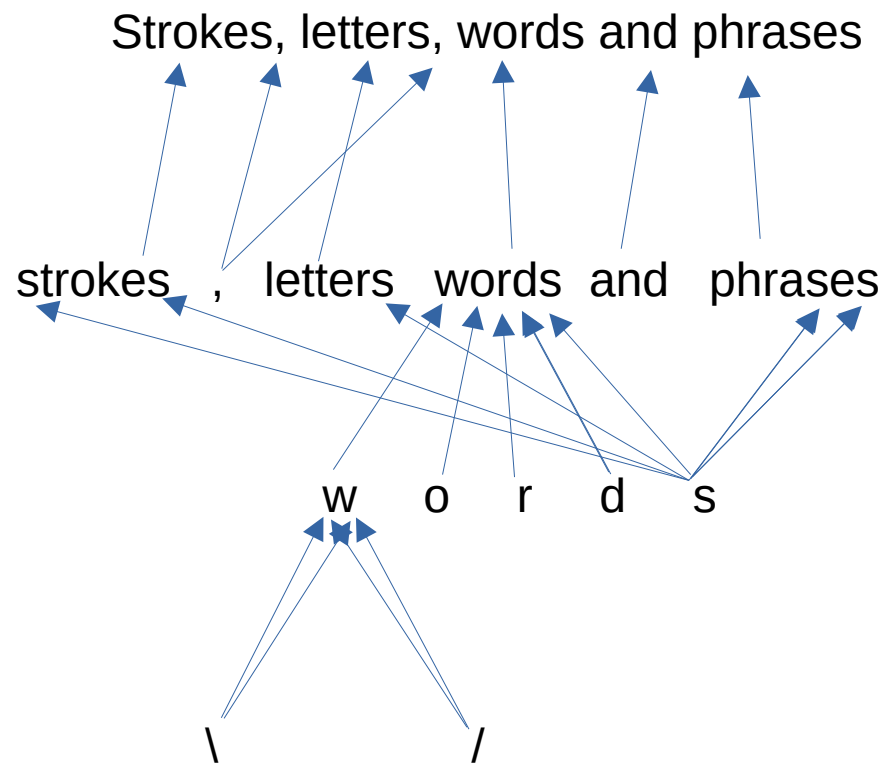
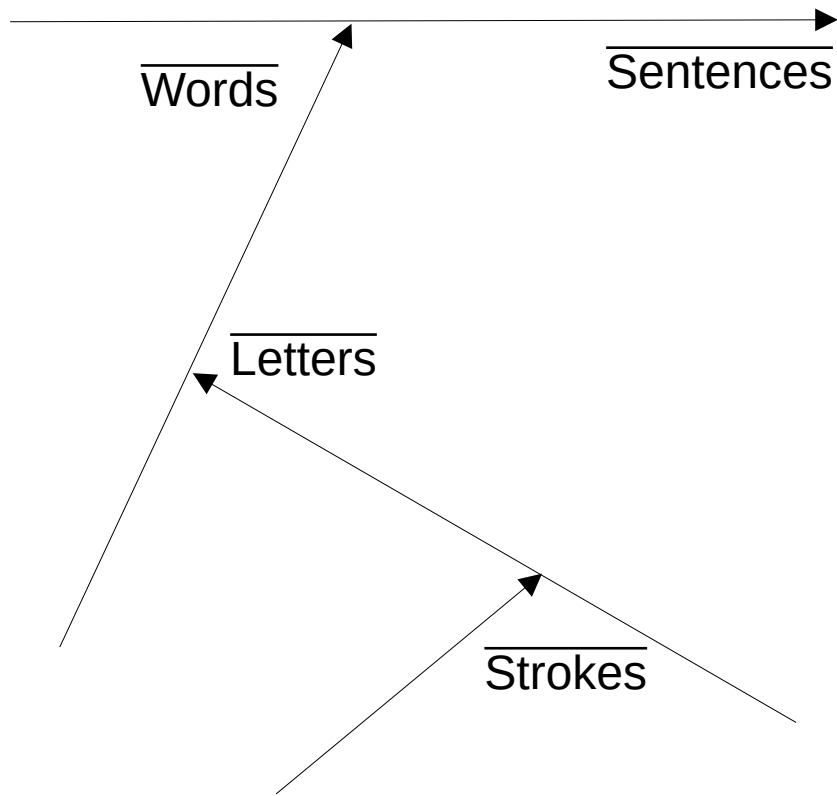
Satisfied needs



Decided & committed actions



Functional equivalence of ~~neural network~~ tensor and (hyper)graph models



Pedro Domingos, Tensor Logic: The Language of AI

<https://arxiv.org/pdf/2510.12269>

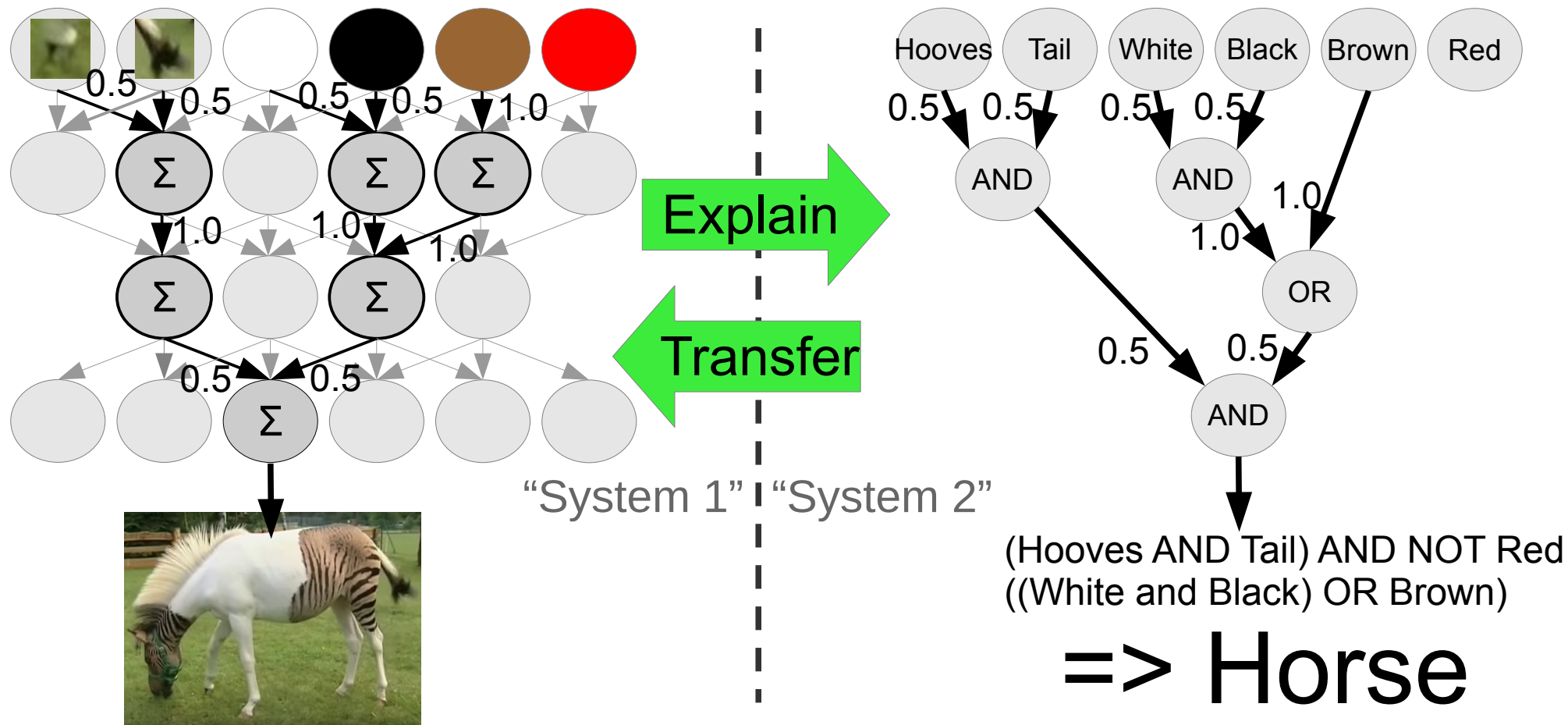
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Graph vertex - dimension

Hyper-graph edge - vector

Hyper-graph - tensor space

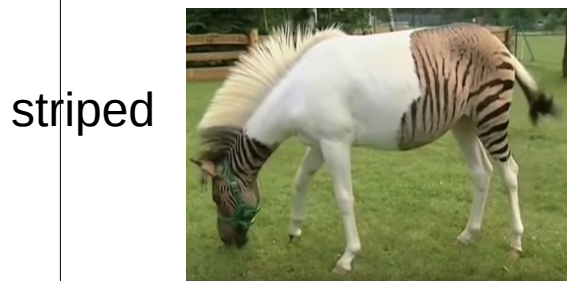
Neuro-Symbolic Integration for Interpretable AI



Typed tensor logic for different kinds of AI-s (logical, sub-symbolic, probabilistic/non-axiomatic)

Truth-Value Tensor
(NARS/PLN/...)

Property **0.0123456**
=750/60750



striped

horse

Subject



Life-long
learning?

Numerical Tensor
(ANN/Bayesian Logic)

Property **~0.01**



striped

horse

Subject

Boolean Tensor
(Boolean Logic)

Property **False**



striped

horse

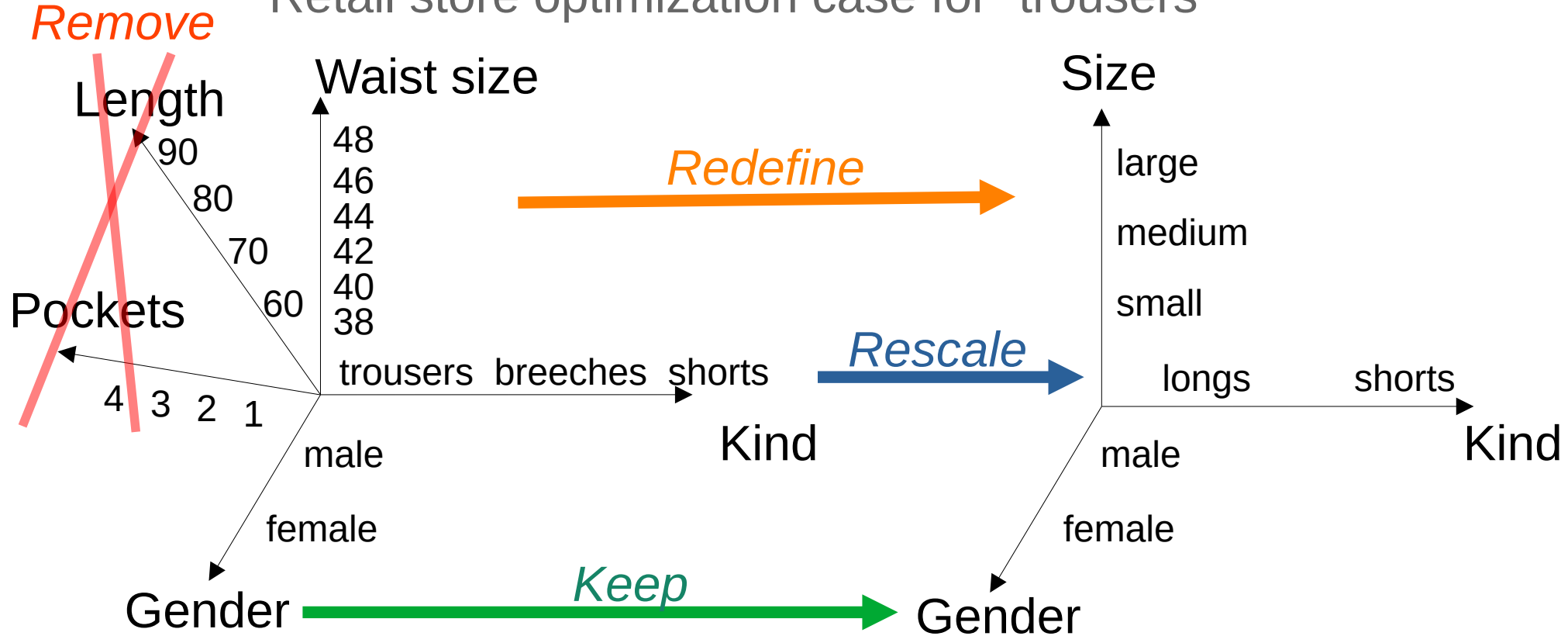
Subject

Pei Wang: Non-Axiomatic Logic
<https://www.worldscientific.com/worldscibooks/10.1142/14486>

Pedro Domingos, Tensor Logic: The Language of AI
<https://arxiv.org/pdf/2510.12269>

Problem of dimensionality (reduction) and discreteness (increase)

Retail store optimization case for “trousers”



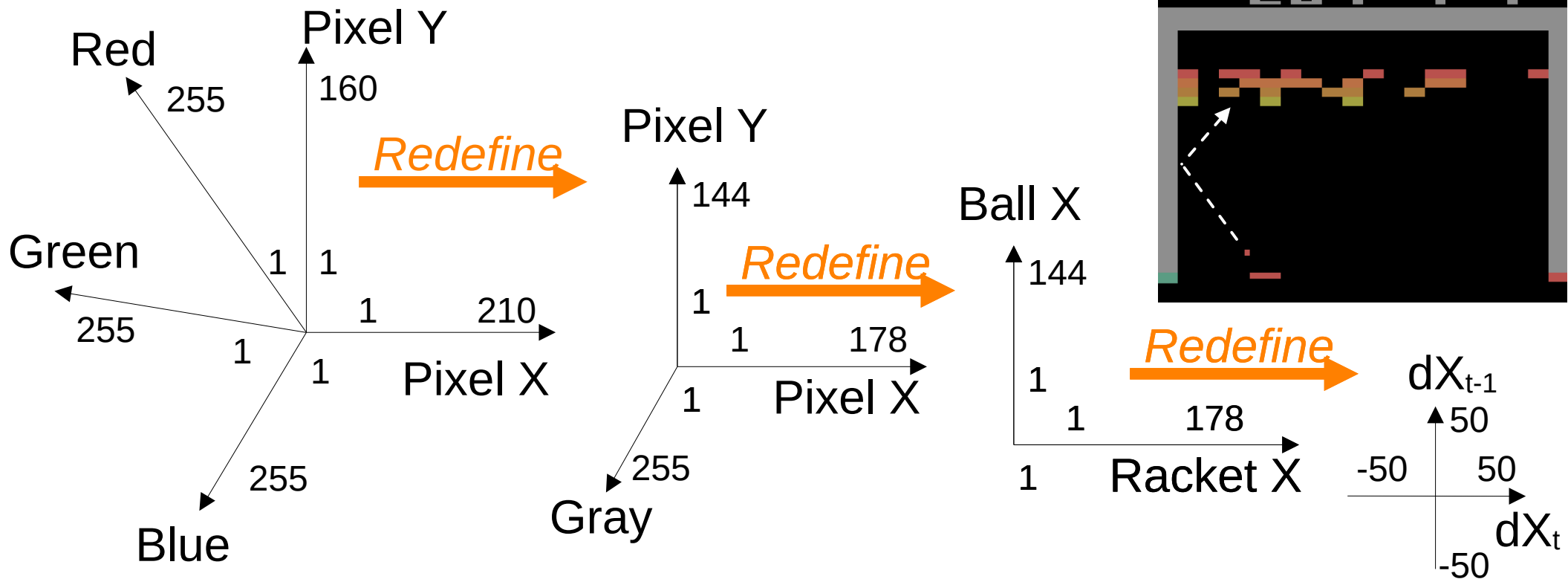
Problem of dimensionality (reduction) and discreteness (increase)

“What You Need is What you See”



Problem of dimensionality (reduction) and discreteness (increase)

Re-defining environment in Atari Breakout



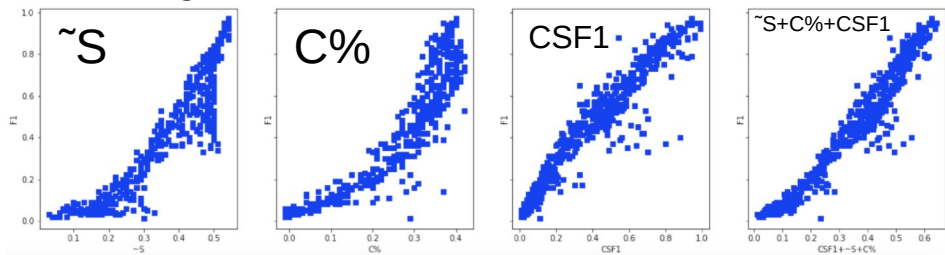
Interpretable representation learning for 3D multi-piece intracellular structures using point clouds

<https://www.nature.com/articles/s41592-025-02729-9>

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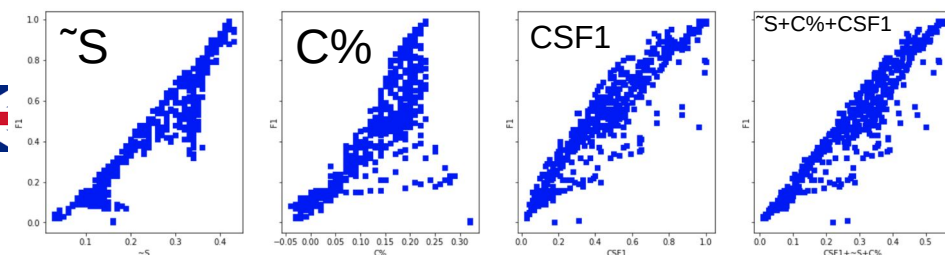
Human language as an evolutionary compression case

English, Train: Brown, Test: Brown 1000

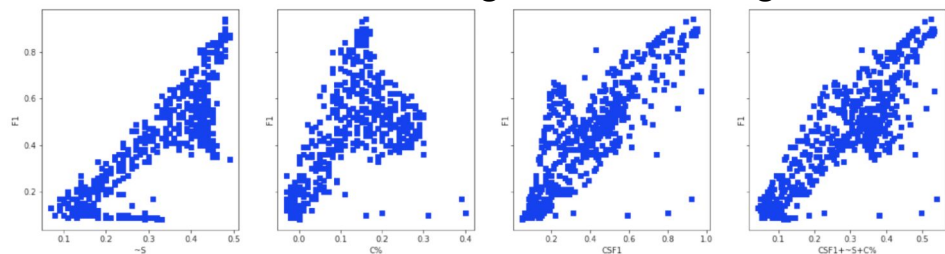


https://github.com/aigents/pygents/blob/main/notebooks/nlp/english/en_tokenizer_auto.ipynb

English, Train: Brown, Test: MagicData 100

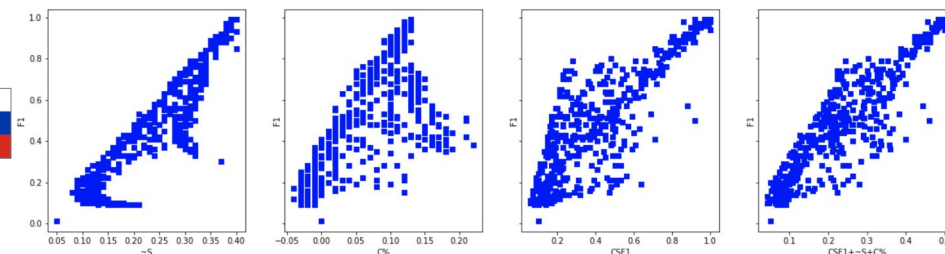


Russian, Train: RusAge, Test: RusAge 1000

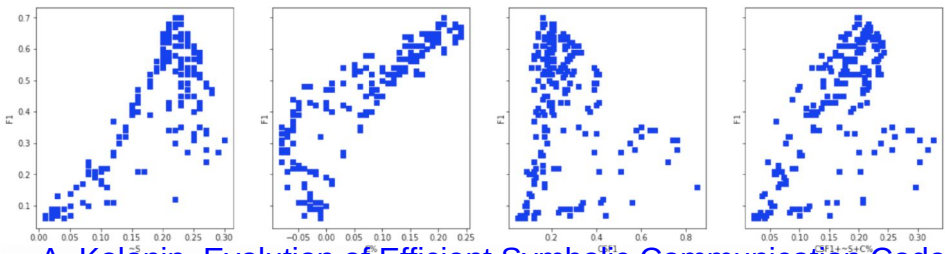


https://github.com/aigents/pygents/blob/main/notebooks/nlp/russian/ru_tokenizer_auto.ipynb

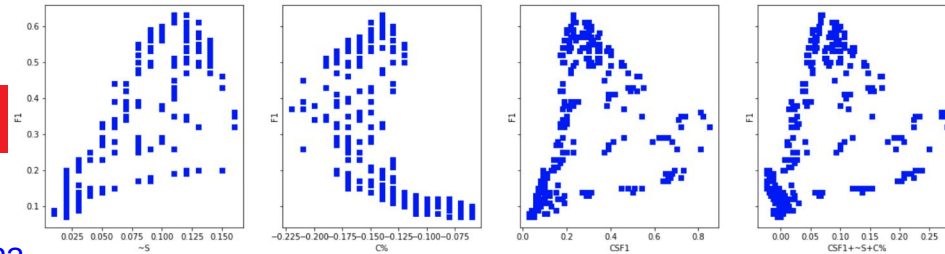
Russian, Train: Brown, Test: MagicData 100



Chinese, Train: CLUE News, Test: CLUE News 1000



Chinese, Train: Brown, Test: MagicData 100



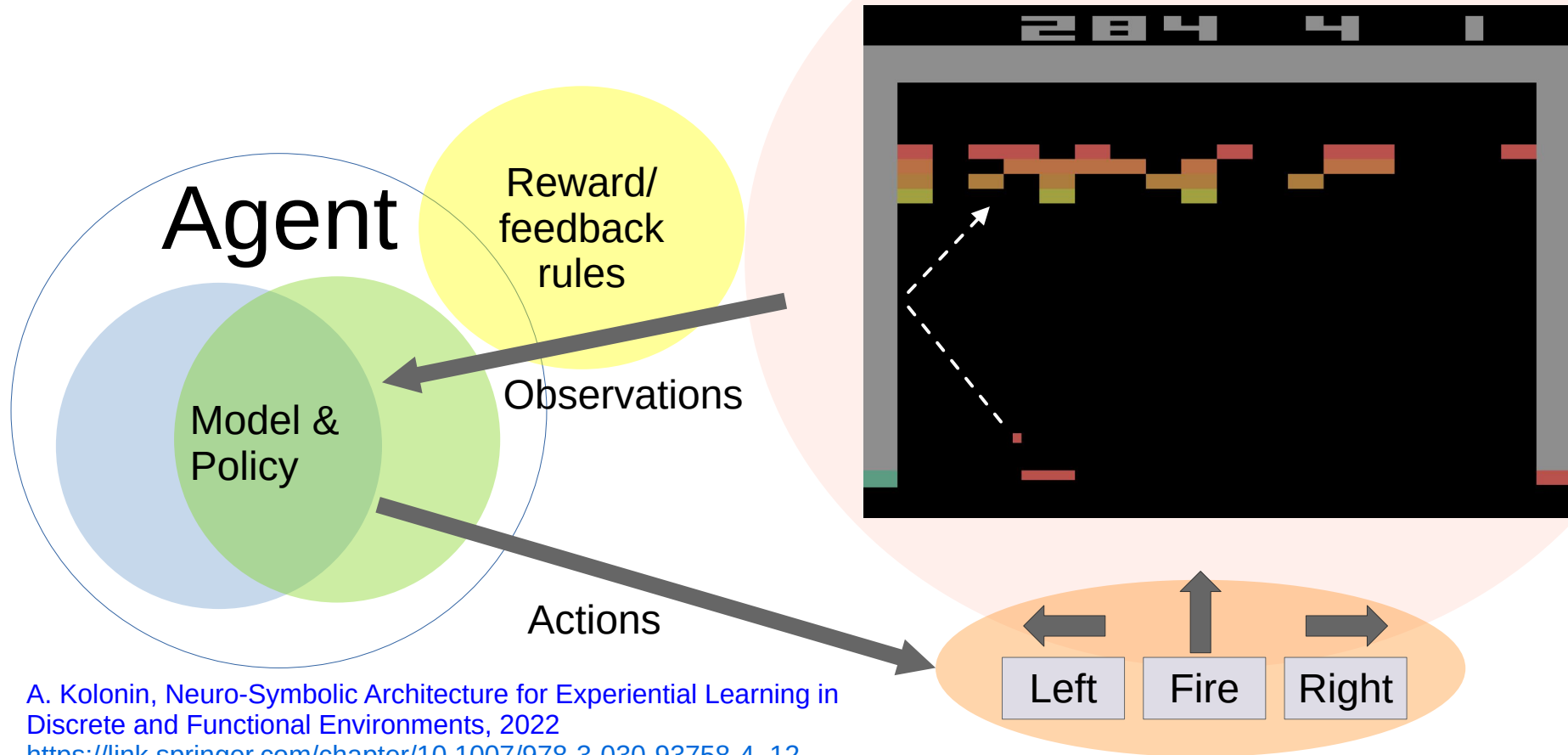
A. Kolonin, Evolution of Efficient Symbolic Communication Codes, 2023

https://link.springer.com/chapter/10.1007/978-3-031-44865-2_1

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Experiential learning case

Virtual gaming environment: OpenAI Gym (Atari Breakout)

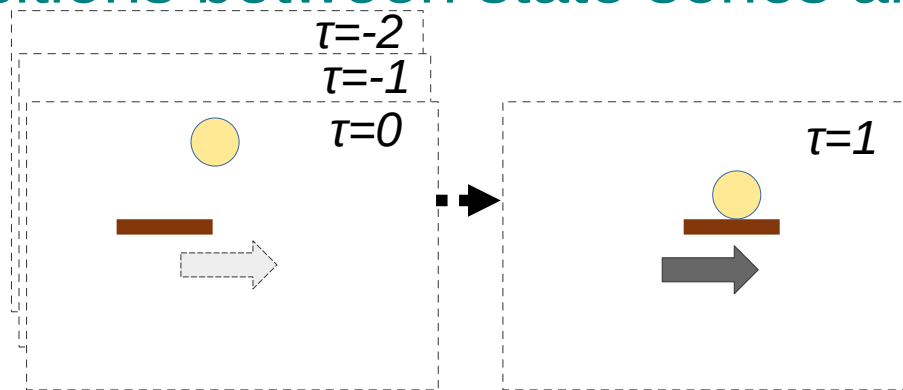


A. Kolonin, Neuro-Symbolic Architecture for Experiential Learning in Discrete and Functional Environments, 2022
https://link.springer.com/chapter/10.1007/978-3-030-93758-4_12

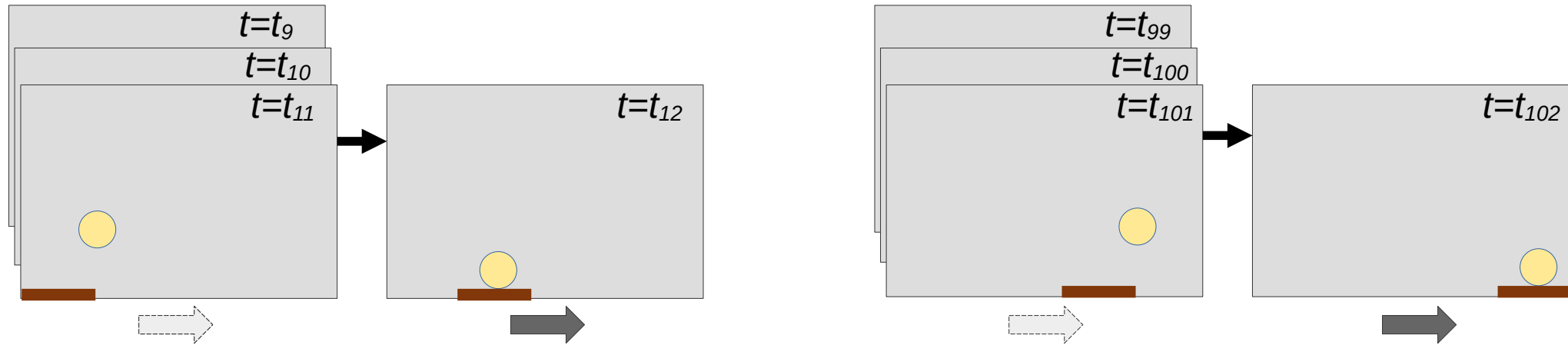
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Experiential learning and decision making based on transitions between state series and “global feedback”

Invariants



Precedents

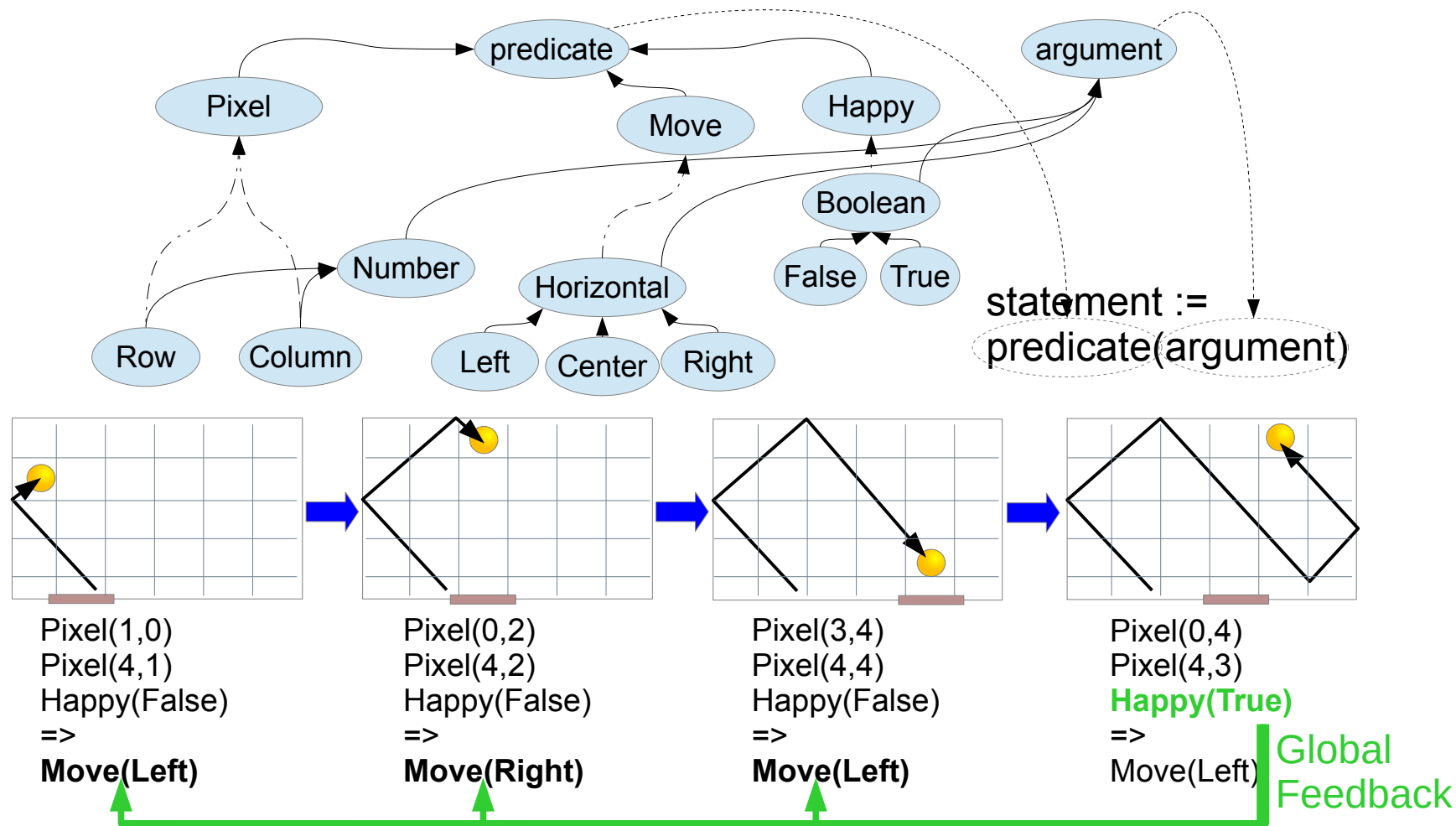


A. Kolonin, Neuro-Symbolic Architecture for Experiential Learning in
Discrete and Functional Environments, 2022

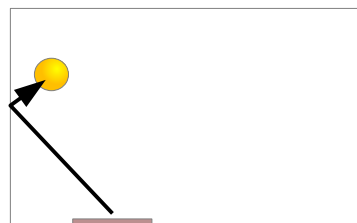
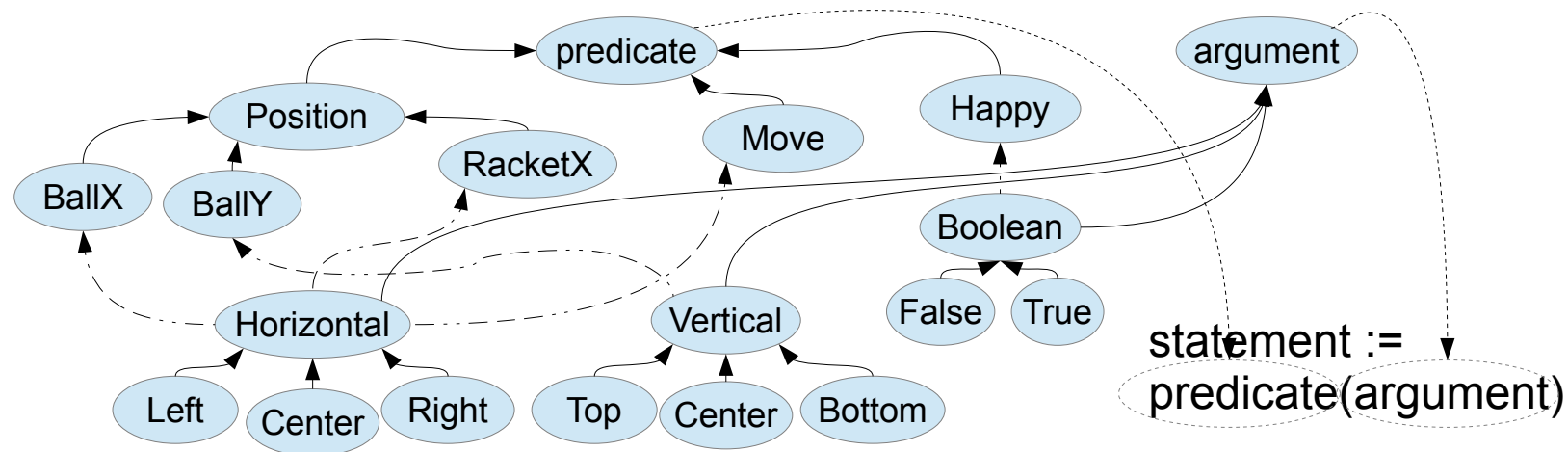
https://link.springer.com/chapter/10.1007/978-3-030-93758-4_12

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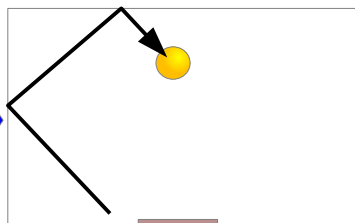
Experiential learning - playing “Pong” (Discrete)



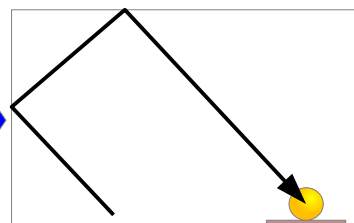
Experiential learning - playing “Pong” (Functional)



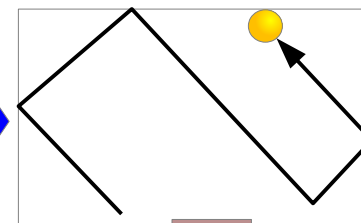
BallY(Top)
BallX(Left)
RacketX(Left)
Happy(False)
=> **Move(Left)**



BallY(Top)
BallX(Center)
RacketX(Center)
Happy(False)
=> **Move(Right)**



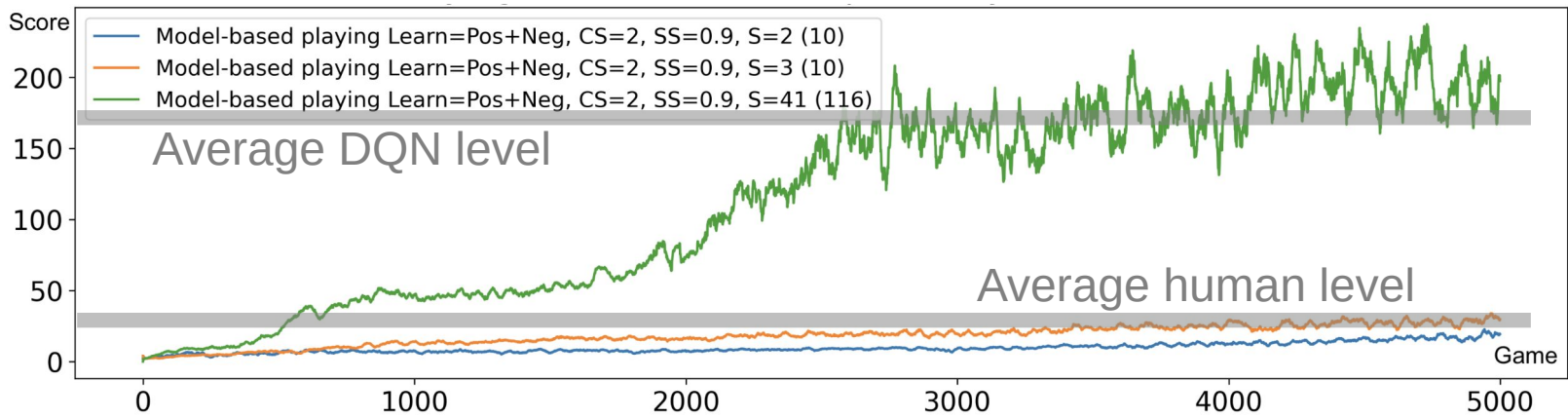
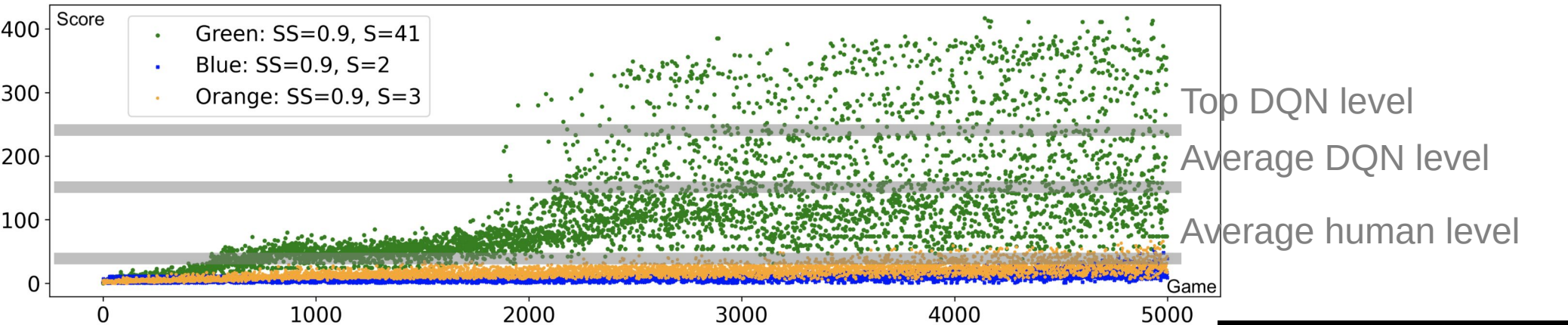
BallY(Bottom)
BallX(Right)
RacketX(Right)
Happy(False)
=> **Move(Right)**



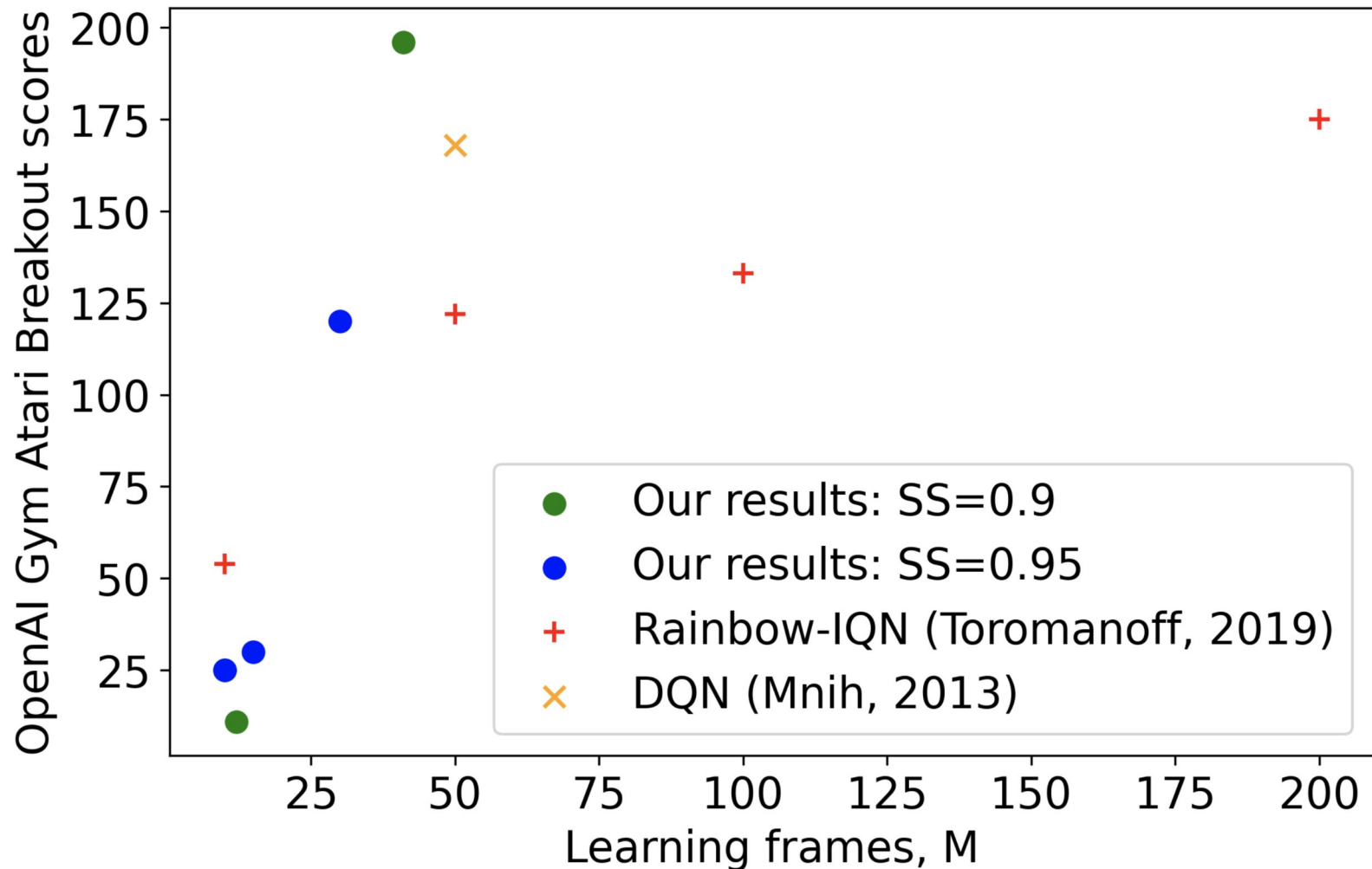
BallY(Bottom)
BallX(Right)
RacketX(Right)
Happy(True)
=> **Move(Left)**

Global
Feedback

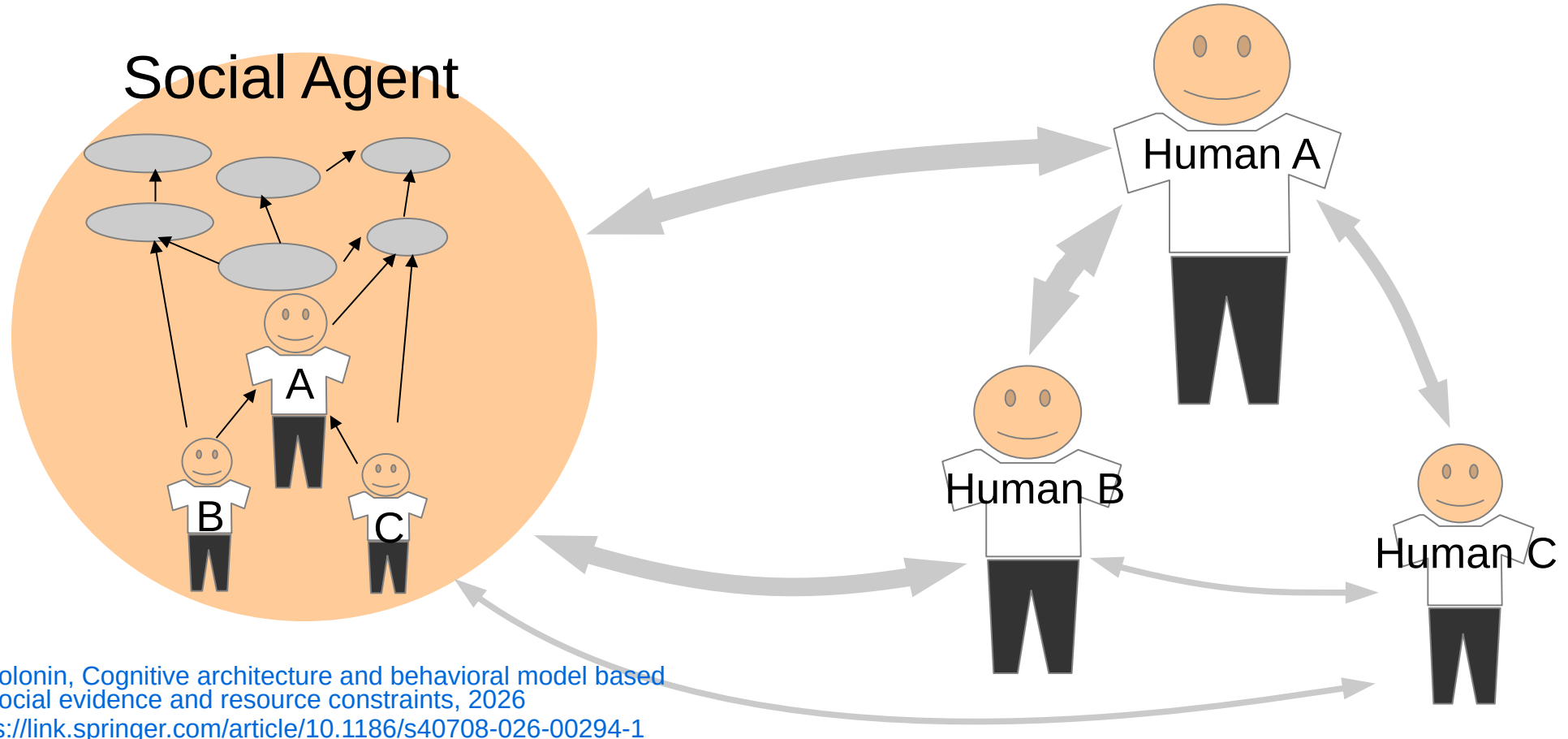
Reinforcement learning – experiential learning and decision making based and transitions between state series and “global feedback”



Reinforcement/experiential learning – comparing to “baselines”

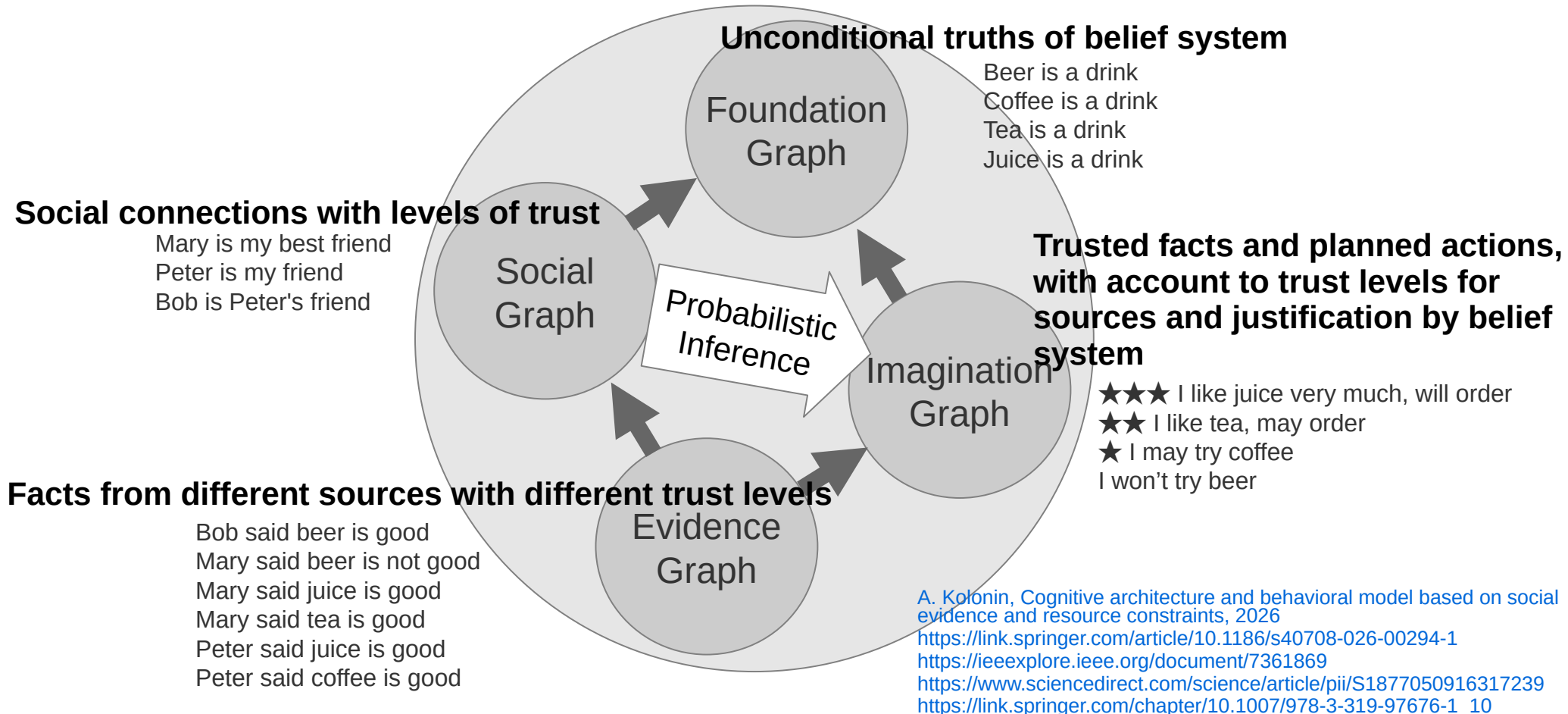


Cognitive architecture and behavioral model based on social evidence and resource constraints



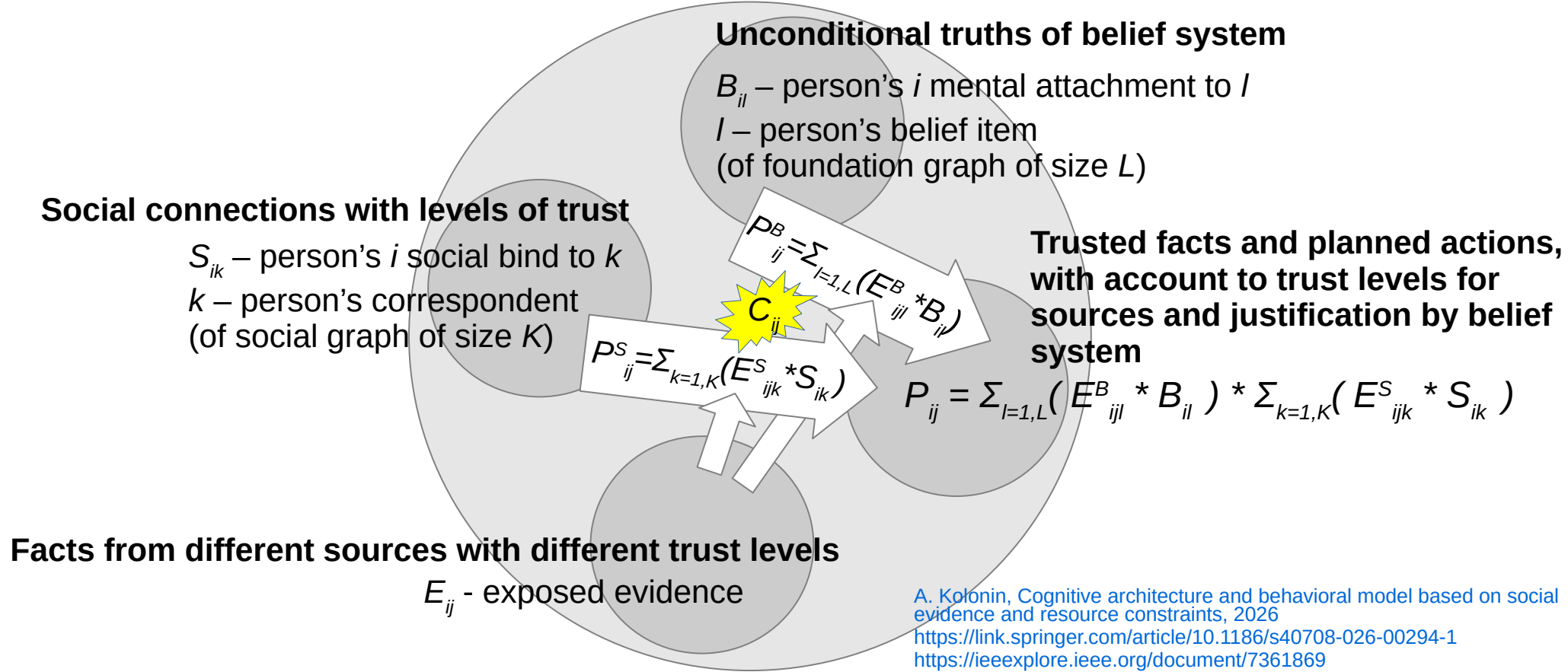
A. Kolonin, Cognitive architecture and behavioral model based on social evidence and resource constraints, 2026
<https://link.springer.com/article/10.1186/s40708-026-00294-1>

Cognitive architecture and behavioral model based on social evidence and resource constraints



A. Kolonin, Cognitive architecture and behavioral model based on social evidence and resource constraints, 2026
<https://link.springer.com/article/10.1186/s40708-026-00294-1>
<https://ieeexplore.ieee.org/document/7361869>
<https://www.sciencedirect.com/science/article/pii/S1877050916317239>
https://link.springer.com/chapter/10.1007/978-3-319-97676-1_10

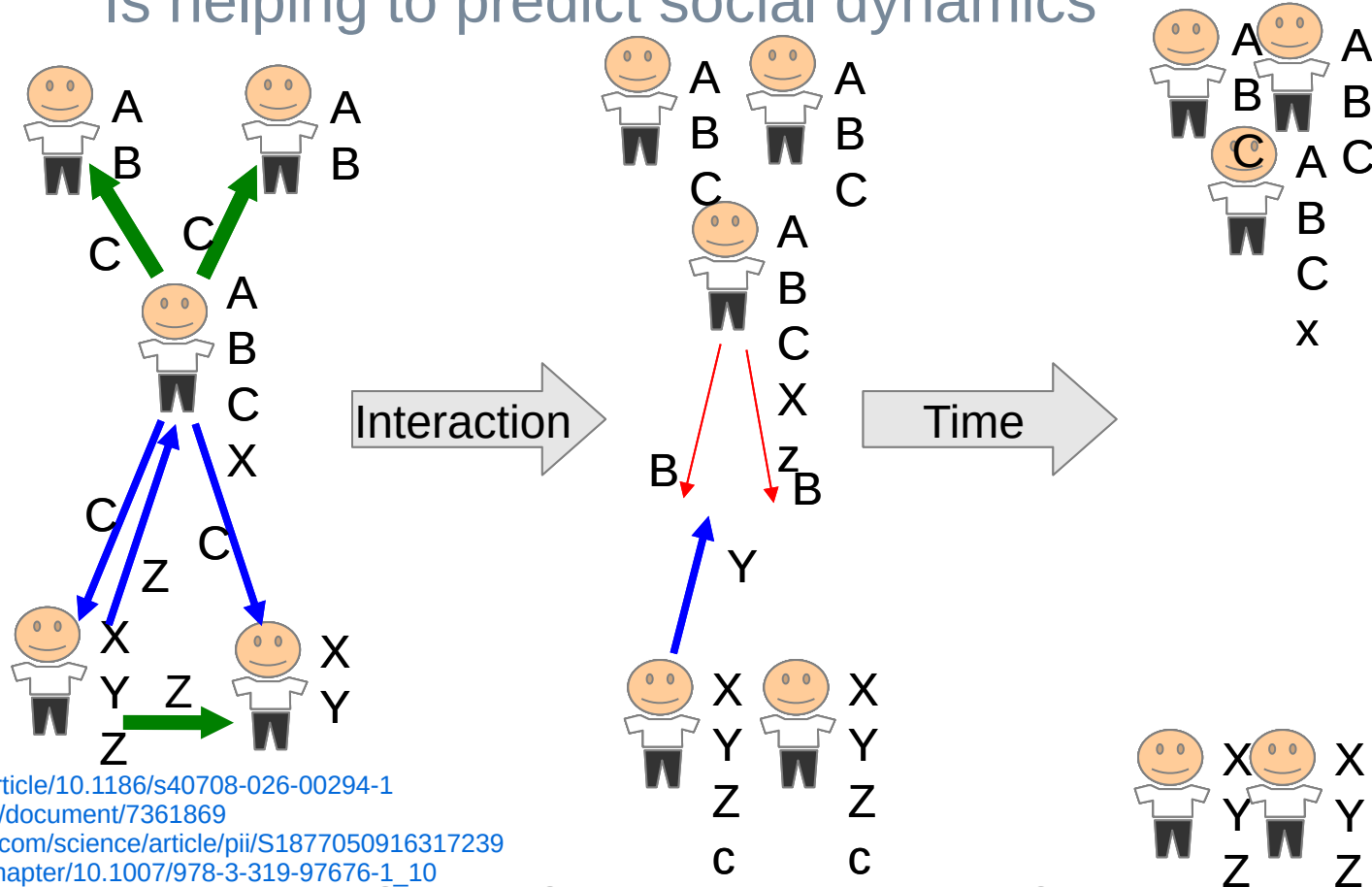
Cognitive architecture and behavioral model based on social evidence and resource constraints



A. Kolonin, Cognitive architecture and behavioral model based on social evidence and resource constraints, 2026
<https://link.springer.com/article/10.1186/s40708-026-00294-1>
<https://ieeexplore.ieee.org/document/7361869>
<https://www.sciencedirect.com/science/article/pii/S1877050916317239>
https://link.springer.com/chapter/10.1007/978-3-319-97676-1_10

Cognitive architecture and behavioral model based on social evidence and resource constraints

Is helping to predict social dynamics



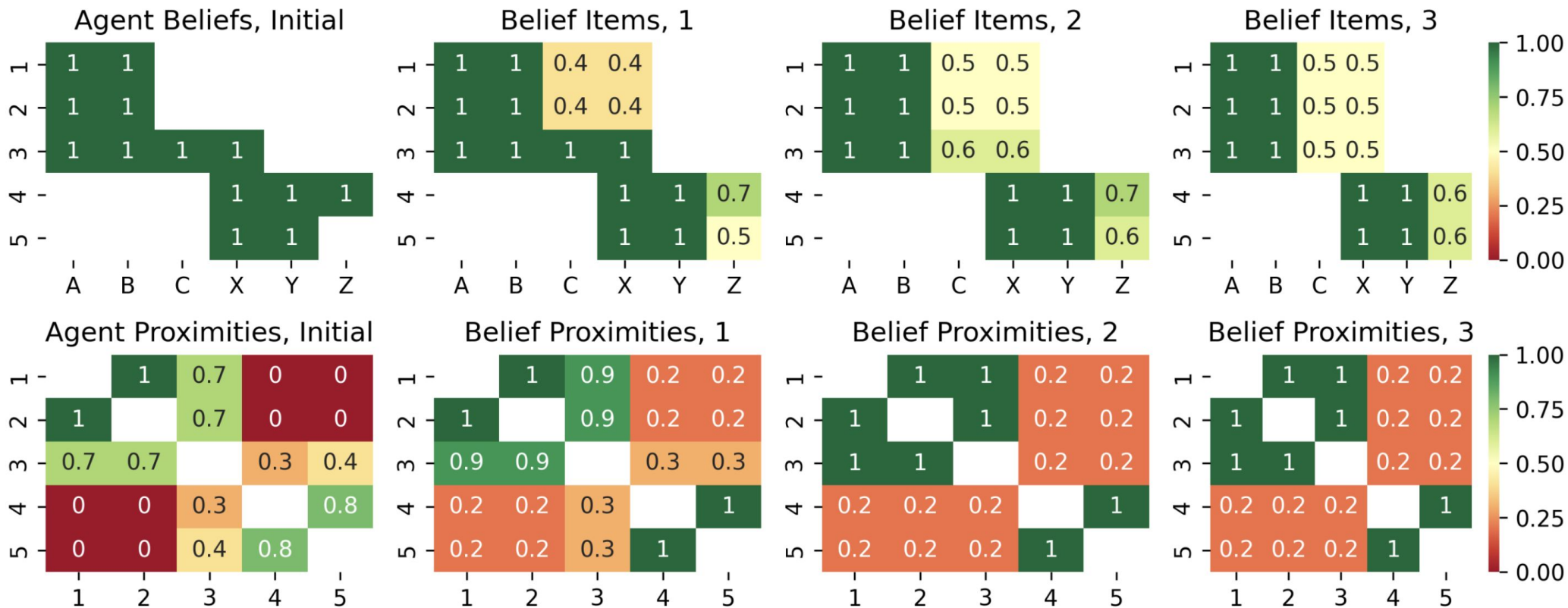
<https://link.springer.com/article/10.1186/s40708-026-00294-1>

<https://ieeexplore.ieee.org/document/7361869>

<https://www.sciencedirect.com/science/article/pii/S1877050916317239>

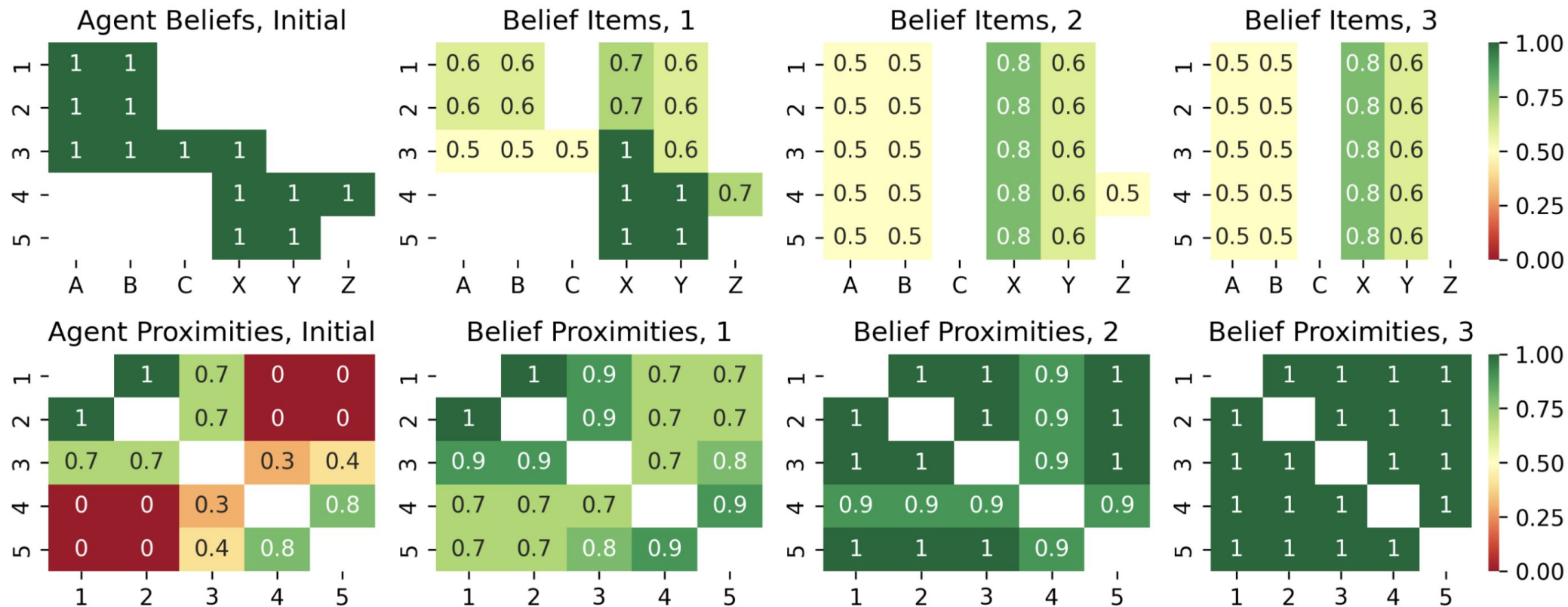
https://link.springer.com/chapter/10.1007/978-3-319-97676-1_10

Simulation: socially limited, cognitively capable community



Results of three rounds of multi-agent simulation with the forgetting threshold peer threshold = 0.5, the social relationship threshold forgetting threshold = 0.0 (limiting social connections K without constraint on belief capacity L). Top row: values of belief items A, B, C, X, Y, Z for five agents 1, 2, 3, 4, 5. Bottom row: social proximity matrices between agents, estimated as the cosine similarity of their beliefs. From left to right: the initial state before the simulation, and then the updated states of beliefs and proximities after three subsequent rounds of multi-agent communication.

Simulation: socially capable, cognitively limited community



Results of three rounds of multi-agent simulation with the forgetting threshold forgetting threshold = 0.5, the social relationship threshold peer threshold = 0.0 (limiting belief capacity L without constraint on social connections K). Top row: values of belief items A, B, C, X, Y, Z for five agents 1, 2, 3, 4, 5. Bottom row: social proximity matrices between agents, estimated as the cosine similarity of their beliefs. From left to right: the initial state before the simulation, and then the updated states of beliefs and proximities after three subsequent rounds of multi-agent communication.

Thank you for attention! Questions?

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Recording from
workshop on the
subject



Paper preprint from
Neuroinformatics-2025
conference

